PARISIAN SIMULTANEOUS RUIN OF BRANCHING BROWNIAN MOTION

ENKELEJD HASHORVA, GORAN POPIVODA, SVYATOSLAV NOVIKOV, AND TIMOFEI SHASHKOV

Abstract. We study simultaneous ruin of a branching Brownian population above the boundary $u + ct$. Every particle then alive must exceed the boundary at each time in a common window or at each of $m + 1$ consecutive grid times. Each split produces between two and K offspring. Sibling correlation matrices may vary between splits and may be singular. For fixed window duration and observation rule, the ruin probability as $u \to \infty$ is asymptotic to $C \exp(-(c + \sqrt{c^2 + 2\beta})u)$, where β is the splitting rate. A double sum over blocks of window starts identifies C in terms of the paths on either side of the first split. Grid observation introduces an average over the position of the split relative to the grid. Conditionally on ruin, the split time and the window start and completion times have the same Gaussian fluctuation. We derive the dense-grid and zero-delay limits of the constants. For continuous observation with zero delay we identify the nonnegative remainder and show that it decays at a strictly faster exponential rate.

1. Introduction and main result

We study the probability that all particles in a branching Brownian population exceed a high linear boundary throughout one common time window. The population may change during the window and the constraint applies to every particle alive at each time. We consider both continuous observation and observation on a uniform grid.

We partition the possible window starts into deterministic blocks, as in the Gaussian double-sum method of [8, Section 5.1]. A window may straddle the first split or a later birth. Conditioning on the first split and reversing the ancestral path gives one integral for a successful block and for a pair of successful blocks. Its dependence on the distance between the blocks controls the Bonferroni correction.

Fix a boundary slope $c > 0$, a splitting rate $\beta > 0$ and a deterministic environment

$$\begin{aligned} &\mathcal{E} = ((k_i, A_i))_{i\geq 1}, \qquad k_i \in \mathbb{N}, \quad 2 \leq k_i \leq K < \infty, \\ &K \in \mathbb{N}, \qquad A_i \in \mathbb{R}^{k_i \times k_i}, \qquad R_i = A_i A_i^\top, \qquad (R_i)_{jj} = 1 \quad 1 \leq j \leq k_i. \end{aligned} \tag{1.1}$$

The correlation matrices R_i may be singular with off-diagonal entries of either sign. The environment may vary arbitrarily from split to split.

Label particles by finite words with $\varnothing$ denoting the ancestor and $v\ell$ the ℓth child of v. Assign independent lifetimes $\zeta_v \sim \mathrm{Exp}(\beta)$ to all potential particles independently of all Brownian motions below. For a particle born at time b_v write $d_v = b_v + \zeta_v$ for its splitting time and set

$$N(t) = \{v : b_v \leq t < d_v\}, \qquad b_\varnothing = 0, \qquad X_\varnothing(t) = B_0(t), \quad 0 \leq t < d_\varnothing,$$

where $N(t)$ ranges over particles actually born and B_0 is a standard Brownian motion. Let σ_i be the ith split in chronological order, let v_i be its parent and put $\sigma_0 = 0$. At time $\sigma_i = d_{v_i}$, replace v_i by k_i children according to

$$\begin{aligned} &N(\sigma_i) = \big(N(\sigma_i-) \setminus \{v_i\}\big) \cup \{v_i 1, \ldots, v_i k_i\}, \\ &b_{v_i\ell} = \sigma_i, \qquad X_{v_i\ell}(\sigma_i + s) = X_{v_i}(\sigma_i-) + W^i_\ell(s), \quad 0 \leq s < \zeta_{v_i\ell}, \quad 1 \leq \ell \leq k_i. \end{aligned}$$

For each birth event i take a vector $\boldsymbol{B}^i$ of K independent standard Brownian motions. The vectors are independent across events and independent of B_0 and all lifetimes. Using the first

Date: 3rd November 2026.

k_i coordinates at event i, set

$$\boldsymbol{W}^i(t) = A_i(B^i_1(t), \dots, B^i_{k_i}(t))^\top, \qquad \mathrm{Cov}(W^i_j(s), W^i_l(t)) = (R_i)_{jl} \min(s,t).$$

Surviving particles keep their original Brownian coordinates. After j splits the population size and holding rate are

$$n_j = 1 + \sum_{i=1}^{j} (k_i - 1) \le 1 + j(K-1), \qquad \beta n_j, \qquad n_0 = 1.$$

After j splits the n_j residual lifetimes are independent exponentials of rate β, by memorylessness. Thus $|N(t)| = n_j$ on $[\sigma_j, \sigma_{j+1})$, and the successive holding times are independent exponentials of rates βn_j. Since $\sum_{j\ge 0} (\beta n_j)^{-1} = \infty$, the pure birth process is nonexplosive. In particular, only finitely many splits occur on each compact time interval.
Define the drifted lower envelope

$$Y(t) = \min_{v \in N(t)} X_v(t) - ct, \qquad t \ge 0. \tag{1.2}$$

At a split the children start at their parent's position. The minimum therefore has the same value on both sides of the split, and Y is continuous.
For $u \ge 0$ and a delay $h \ge 0$ define simultaneous Parisian ruin by

$$S^{0,h}_u = \inf\left\{ t \ge 0 : \inf_{0 \le r \le h} Y(t+r) > u \right\}, \qquad T^{0,h}_u = S^{0,h}_u + h. \tag{1.3}$$

With $\inf \varnothing = \infty$, put

$$\mathcal{A}_{0,h}(u) = \{S^{0,h}_u < \infty\}, \qquad p_{0,h}(u) = \mathbb{P}\{\mathcal{A}_{0,h}(u)\}.$$

Every reserve $u + ct - X_v(t)$ present during a successful window is strictly negative. Particles continue to evolve after crossing. Since the inequality in (1.3) is strict, $S^{0,h}_u$ is the infimum of the successful starts and need not be a successful start itself.
For a mesh $\delta > 0$ and $m \in \mathbb{N}_0$, set

$$S^{\delta,m}_u = \delta \inf\left\{ j \in \mathbb{N}_0 : \min_{\ell = 0,\dots,m} Y((j+\ell)\delta) > u \right\}, \qquad T^{\delta,m}_u = S^{\delta,m}_u + m\delta, \tag{1.4}$$

and

$$\mathcal{A}_{\delta,m}(u) = \{S^{\delta,m}_u < \infty\}, \qquad p_{\delta,m}(u) = \mathbb{P}\{\mathcal{A}_{\delta,m}(u)\}.$$

This requires $m+1$ consecutive observations over duration $m\delta$ and imposes no condition between observations.
Taking $k_i \equiv k$ and $R_i \equiv I_k$ gives the usual k-ary branching Brownian motion (BBM). Zero delay ($h = 0$ or $m = 0$) gives ordinary simultaneous ruin under the chosen observation rule. For finite-horizon simultaneous exceedances in Brownian decision trees with prescribed split times see [10]. For absorption at a barrier and the all-time minimum of binary BBM with drift see [2].
For Parisian ruin in spectrally negative Lévy and discrete-time risk models see, respectively, [5, 17] and [6, 19]. Parisian ruin in Gaussian models is treated in [1, 8, 9, 14, 18]. For Gaussian claims with independent random premium income see [20], and for discrete observation in Brownian risk models see [15].
With branching suppressed, the envelope reduces to $B_0(t) - ct$. We assume $\beta > 0$ throughout. Set

$$a = \sqrt{c^2 + 2\beta}, \qquad \kappa = c + a, \qquad s_u = \sqrt{u/a^3}. \tag{1.5}$$

Let $\tau = \sigma_1$ be the ancestor's split time. Start its cohort of k_1 siblings at zero with covariance R_1 and continue the environment from its second entry. For this cohort define

$$L(s) = \min_{v \text{ alive in this cohort at } s} X_v(s) - cs, \qquad M = \sup_{s \ge 0} L(s), \qquad H(x) = \mathbb{P}\{M > x\}, \quad x \ge 0. \tag{1.6}$$

The cohort is independent of (τ, B_0). Let W be an independent standard Brownian motion and define the continuous two-sided field

$$Z(s) = \begin{cases} W(-s) - a(-s), & s \le 0, \\ L(s), & s \ge 0. \end{cases} \tag{1.7}$$

For $h \ge 0$ and $0 \le \theta < \delta$ put

$$\Gamma_h(Z) = \sup_{s\in\mathbb{R}} \inf_{0\le r\le h} Z(s+r), \tag{1.8}$$

$$V_{\delta,m,\theta}(Z) = \sup_{j\in\mathbb{Z}} \min_{\ell=0,\dots,m} Z((j+\ell)\delta - \theta). \tag{1.9}$$

Time zero in Z corresponds to the first split. Windows may straddle zero, and the functionals in (1.8) and (1.9) can be negative. The expectations below include both the Brownian paths and the genealogy. We suppress their dependence on $\mathcal{E}$.

Theorem 1.1. *Consider the branching model above with $c, \beta > 0$ and the bounded offspring and covariance sequence satisfying (1.1). For fixed $h \ge 0$, $\delta > 0$ and $m \in \mathbb{N}_0$, we have as $u \to \infty$*

$$p_{0,h}(u) \sim C_{0,h} \exp(-\kappa u), \qquad p_{\delta,m}(u) \sim C_{\delta,m} \exp(-\kappa u), \tag{1.10}$$

where

$$C_{0,h} = \frac{\beta}{a\kappa} \mathbb{E}\{\exp(\kappa \Gamma_h(Z))\} \in (0, \infty), \tag{1.11}$$

$$C_{\delta,m} = \frac{\beta}{a\kappa\delta} \int_0^\delta \mathbb{E}\{\exp(\kappa V_{\delta,m,\theta}(Z))\} \mathrm{d}\theta \in (0, \infty). \tag{1.12}$$

The offspring numbers and sibling correlations enter the constants through the law of L in (1.7).

2. Discussions and Extensions

Corollary 2.1. *Under the assumptions of Theorem 1.1, fix either observation rule. Write $\mathcal{A}(u)$ for its ruin event, S_u, T_u for its window start and completion times as defined in (1.3) or (1.4), and C for its constant. If G is standard normal, then*

$$\left(\frac{\tau - u/a}{s_u}, \frac{S_u - u/a}{s_u}, \frac{T_u - u/a}{s_u} \right) \Bigg| \, \mathcal{A}(u) \quad \underset{u\to\infty}{\Longrightarrow} \quad (G, G, G). \tag{2.1}$$

Since Φ, the standard normal distribution function, is continuous, the convergence of the marginal distribution functions is uniform. Consequently, for $U_u \in \{S_u, T_u\}$,

$$\sup_{t\ge 0} \left| \exp(\kappa u) \mathbb{P}\{U_u \le t\} - C\Phi\left(\frac{t - u/a}{s_u} \right) \right| \xrightarrow[u\to\infty]{} 0. \tag{2.2}$$

Crossing before the first split yields a direct lower bound. Put $\vartheta_u = \inf\{t \ge 0 : B_0(t) - ct > u\}$. Independence of τ and B_0 and the Brownian hitting-time transform [4] yield

$$p_{0,0}(u) \ge \mathbb{P}\{\vartheta_u < \tau\} = \mathbb{E}\{\exp(-\beta\vartheta_u)\} = \exp(-\kappa u), \qquad u \ge 0.$$

The contribution of crossings after the first split can be expressed in terms of the maximum M of the newborn cohort.

Proposition 2.2. *In the setting of Theorem 1.1, let $H(x) = \mathbb{P}\{M > x\}$ as in (1.6). Then*

$$C_{0,0} = 1 + \frac{\beta}{a} \int_0^\infty \big(\exp(\kappa x) - \exp(-(a-c)x)\big) H(x) \mathrm{d}x \tag{2.3}$$

$$= \frac{(a-c)\mathbb{E}\{\exp(\kappa M)\} + \kappa \mathbb{E}\{\exp(-(a-c)M)\}}{2a}. \tag{2.4}$$

For every $u \ge 0$ one has the exact identity

$$C_{0,0} - \exp(\kappa u) p_{0,0}(u) = \frac{\beta}{a} \int_u^\infty \exp(\kappa x)\big(1 - \exp(-2a(x-u))\big) H(x) \mathrm{d}x. \tag{2.5}$$

For every $\kappa < r < c + \sqrt{c^2 + 4\beta}$ the remainder $C_{0,0}\exp(-\kappa u) - p_{0,0}(u)$ is nonnegative and $O(\exp(-ru))$ as $u \to \infty$, uniformly over all admissible environments. Moreover, $C_{0,0} > 1$.

Corollary 2.3. *The map $h \mapsto C_{0,h}$ is continuous on $[0,\infty)$. If $\delta_n \downarrow 0$ and $m_n \in \mathbb{N}_0$ satisfy $m_n\delta_n \to h \geq 0$, then*

$$C_{\delta_n, m_n} \longrightarrow C_{0,h}. \tag{2.6}$$

For ordinary ruin, the constant at any fixed mesh $\delta > 0$ satisfies

$$0 < C_{\delta,0} < C_{0,0}. \tag{2.7}$$

The limit in (2.6) is taken after the high-threshold limit defining each constant. At positive duration $m\delta$, a dip between observations can invalidate a continuous window. Conversely, an off-grid interval of length $m\delta$ contains only m grid points.

Remark (Long windows and coarse grids). The window minimum is nonincreasing as h grows, or as m grows at fixed δ. Taking the supremum over starts preserves this ordering. In either case a window of duration d has an endpoint at distance at least $d/2$ from zero. By (A.4) its exponentiated minimum is at most $A_\varepsilon \exp(-\kappa\varepsilon d/2)$, uniformly in the start. Hence

$$C_{0,h} \downarrow 0 \quad (h \to \infty), \qquad C_{\delta,m} \downarrow 0 \quad (m \to \infty,\ \delta \text{ fixed}). \tag{2.8}$$

For $m = 0$, bounding the phase-dependent maximum in (1.12) by the sum over all grid sites gives

$$\delta C_{\delta,0} \leq \frac{\beta}{a\kappa} \int_{\mathbb{R}} \mathbb{E}\{\exp(\kappa Z(s))\}\mathrm{d}s < \infty.$$

The integral is finite by (A.4) and independent of δ, so $C_{\delta,0} \to 0$ as $\delta \to \infty$. Finally, for a positive integer q, take Θ uniform on $[0, q\delta)$ independently of Z. The lattice $q\delta\mathbb{Z} - \Theta$ is contained in $\delta\mathbb{Z} - \Theta$, and $\Theta \bmod \delta$ is uniform on $[0,\delta)$. This gives $C_{q\delta,0} \leq C_{\delta,0}$.

The extension to random or generation-indexed environments follows the localisation estimates in Section A.

3. Proofs

Proof of Theorem 1.1. Write $y_u(t) = (t - u/a)/s_u$. First take a continuous $f \geq 0$ with support in $[-\Lambda + 1, \Lambda - 1]$ for some $\Lambda > 1$. Define

$$\mu_u(B) = \exp(\kappa u)\mathbb{E}\{f(y_u(\tau))\mathbf{1}_B\}.$$

Fix a block length $T > 0$. In continuous time take $D = [0,h]$, $D_T = [0,T)$. On the grid take $T = N\delta$ with $N \geq 1$ an integer, $D = \{0,\delta,\dots,m\delta\}$ and $D_T = \{0,\delta,\dots,(N-1)\delta\}$. Denote success at some start in $b + D_T$ by A_b, where $b = jT$, $j \in \mathbb{N}_0$. Set

$$q(s) = \exp\left(\kappa \inf_{d\in D} Z(s+d)\right), \qquad Q_T(z) = \sup_{v\in D_T} q(z+v), \qquad \mathcal{H}_T = \int_{\mathbb{R}} \mathbb{E}\{Q_T(z)\}\mathrm{d}z.$$

Since $0 \in D$, (A.4) gives, with $\alpha = \kappa\varepsilon$,

$$q(s) \leq A_\varepsilon \exp(-\alpha|s|), \qquad Q_T(z) \leq A_\varepsilon \exp\big(-\alpha \operatorname{dist}(0,[z,z+T])\big).$$

One and two blocks. Take $(B, K(z)) = (A_b, Q_T(z))$ for one block. For two blocks at separation nT, $n \geq 1$, take

$$B = A_b \cap A_{b+nT}, \qquad K(z) = \min\{Q_T(z), Q_T(z+nT)\}.$$

Condition on $\tau = t$. Relative to the split position, the reversed path $Y(t-v) - Y(t)$, $0 \leq v \leq t$, is Brownian motion with drift c, independent of the forward cohort. Its density relative to $Z(-v) = W(v) - av$ on $[0,t]$ is $\exp(\kappa Z(-t) + \beta t)$ by the Brownian change-of-drift formula [16], since $\kappa = a + c$ and $a^2 - c^2 = 2\beta$. Multiplication by the split density $\beta\exp(-\beta t)$ leaves $\beta\exp(\kappa Z(-t))$. Since $Y(0) = 0$, the absolute field at time $t+s$ is represented by $Z(s) - Z(-t)$

for $s \geq -t$. Put $z = b - t$ and $x = u + Z(-t)$. Then B becomes $\{K(z) > \exp(\kappa x)\}$, and the weight in μ_u times the tilted split density is $\beta f(y_u(t)) \exp(\kappa x) \mathrm{d}t$.
Let $r = (-z)_+$. The block variable $K(z)$ is determined by L and $(W(v))_{0 \leq v \leq r}$. Under the reference law $x = u - at + W(r) + (W(t) - W(r))$, with the last increment independent of these paths. For $b > 0$ the conditional density of x is therefore $\phi_{t-r}(x - u + at - W(r))$, where $t - r = \min(t, b) > 0$ and ϕ_v is the centred normal density with variance v. Introduce an independent $\xi \sim \mathrm{Exp}(\kappa)$. For $k > 0$ and $g \geq 0$,

$$\int_{-\infty}^{\kappa^{-1} \log k} \exp(\kappa x) g(x) \mathrm{d}x = \frac{k}{\kappa} \mathbb{E}\big\{ g(\kappa^{-1} \log k - \xi) \big\}.$$

Apply this identity with $k = K(z)$ and set $\Delta(z) = \kappa^{-1} \log K(z) - \xi - W(r)$. With $r = (t-b)_+$, for $b > 0$ we obtain

$$\mu_u(B) = \frac{\beta}{\kappa} \int_0^\infty f(y_u(t)) \, \mathbb{E}\{K(b-t) \phi_{t-r}(at - u + \Delta(b-t))\} \mathrm{d}t. \tag{3.1}$$

For a single block let $y_u(b) \to y \in \mathbb{R}$ and fix T. Change variables to $z = b - t$ in (3.1). At fixed z, for large u we have $z < b$, $t = b - z \sim u/a$ and $y_u(t) \to y$. In particular, $s_u / \sqrt{t - r} \to 1/a$ and $(at - u)/\sqrt{t - r} \to y$. The random shift $\Delta(z)$ is finite and independent of u. Thus $s_u \phi_{t-r}(at - u + \Delta(z)) \to \phi(y)/a$ almost surely, where $\phi = \phi_1$. Whenever $f(y_u(t)) \neq 0$, we have $t \geq u/(2a)$ for all sufficiently large u. The same lower bound holds for b since $y_u(b)$ is bounded. Hence $s_u \phi_{t-r} \leq 1/(a\sqrt{\pi})$ wherever the weight is nonzero. The resulting integrable majorant is a constant times $\mathbb{E}\{Q_T(z)\}$, so

$$s_u \mu_u(A_b) \longrightarrow \frac{\beta \mathcal{H}_T}{a\kappa} f(y) \phi(y). \tag{3.2}$$

Since this holds along every sequence $y_u(b) \to y$, the approximation by the right-hand side at $y = y_u(b)$ is uniform on compact sets.
For a pair and $|y_u(b)| \leq \Lambda$, apply the same density bound in (3.1) for all sufficiently large u. The larger of the distances from zero to $[z, z+T]$ and $[z + nT, z + (n+1)T]$ is $(n-1)T/2 + |z + (n+1)T/2|$. Using the bound on Q_T for the farther interval gives, uniformly in $n \geq 1$,

$$\begin{aligned} s_u \mu_u(A_b \cap A_{b+nT}) &\leq \frac{\beta \|f\|_\infty}{a\kappa\sqrt{\pi}} \int_{\mathbb{R}} \mathbb{E}\{K(z)\} \mathrm{d}z \\ &\leq C_1 \|f\|_\infty \exp(-\alpha(n-1)T/2), \end{aligned} \tag{3.3}$$

where C_1 is independent of n, T, Λ.
The double sum. Let $I_u = \{j \in \mathbb{N}_0 : |y_u(jT)| \leq \Lambda\}$ and $U_u = \bigcup_{j \in I_u} A_{jT}$. Write P_1 for the sum of $\mu_u(A_{jT})$ over I_u and P_2 for the sum over unordered pairwise intersections. Then $P_1 - P_2 \leq \mu_u(U_u) \leq P_1$. By (3.2) the single sum is a Riemann sum with mesh T/s_u, and

$$P_1 \longrightarrow \frac{\beta \mathcal{H}_T}{a\kappa T} \int_{\mathbb{R}} f(y) \phi(y) \mathrm{d}y.$$

Each separation nT occurs in at most $1 + 2\Lambda s_u / T$ pairs. Summing (3.3) over $n \geq 1$ gives

$$\limsup_{u \to \infty} P_2 \leq \frac{2C_1 \Lambda \|f\|_\infty}{T(1 - \exp(-\alpha T/2))}.$$

On $\{f(y_u(\tau)) \neq 0\}$ an omitted block starts more than s_u from τ. Success in that block gives a crossing more than $s_u - T$ from τ. For fixed T, R, the event $E_{s_u - T}(u)$ of Lemma A.3 is contained in $E_R(u)$ for all sufficiently large u. Taking $u \to \infty$ and then $R \to \infty$ in (A.8) yields $\mu_u(\mathcal{A}(u) \setminus U_u) \to 0$.
The limit of the block constant. In either observation scheme the part of $\mathcal{H}_T$ outside $[-T, 0]$ is at most $2\mathbb{E}\{A_\varepsilon\}/\alpha$. In continuous time, $Q_T(z) \leq \sup_s q(s)$. Conversely, for each path and each s, the set of z with $s \in z + D_T$ has length T. Thus $\int_{\mathbb{R}} Q_T(z) \mathrm{d}z \geq T \sup_s q(s)$ pathwise, and

$$0 \leq \mathcal{H}_T - T\mathbb{E}\{\exp(\kappa \Gamma_h(Z))\} \leq 2\mathbb{E}\{A_\varepsilon\}/\alpha.$$

On the grid write $q_i(\theta) = q(i\delta - \theta)$ and $Q(\theta) = \sup_i q_i(\theta) = \exp(\kappa V_{\delta,m,\theta}(Z))$. Decomposing $z = k\delta - \theta$ gives

$$\mathcal{H}_{N\delta} = \int_0^\delta \mathbb{E}\left\{ \sum_{k\in\mathbb{Z}} \max_{0\le j<N,\ j\in\mathbb{Z}} q_{k+j}(\theta) \right\} \mathrm{d}\theta.$$

Each index i belongs to N blocks, so the sum is at least $NQ(\theta)$ pathwise. On $[-N\delta, 0]$ the bound $Q_T(z) \le Q((-z) \bmod \delta)$ integrates over N periods. Hence

$$0 \le \mathcal{H}_{N\delta} - N\int_0^\delta \mathbb{E}\{Q(\theta)\}\mathrm{d}\theta \le 2\mathbb{E}\{A_\varepsilon\}/\alpha.$$

Taking $u \to \infty$ and then $T \to \infty$ in the Bonferroni bounds (through multiples of δ for grid observation) now gives

$$\lim_{u\to\infty} \exp(\kappa u)\mathbb{E}\big\{f(y_u(\tau))\mathbf{1}_{\mathcal{A}(u)}\big\} = C\mathbb{E}\{f(G)\}, \tag{3.4}$$

where G is standard normal and C is given by (1.11) or (1.12). For a bounded continuous f, take a continuous cutoff $0 \le \chi_J \le 1$ equal to one on $[-J, J]$ and zero outside $[-J-1, J+1]$. Apply the compact-support result to the positive and negative parts of $f\chi_J$. On removing the cutoff, the error on the left of (3.4) is bounded by $\|f\|_\infty \exp(\kappa u)\mathbb{P}\{\mathcal{A}(u),\ |y_u(\tau)| > J\}$. By (A.6) its limit superior tends to zero as $J \to \infty$. Since $\mathbb{E}\{f(G)\chi_J(G)\} \to \mathbb{E}\{f(G)\}$, this proves (3.4) for all bounded continuous f. Setting $f = 1$ proves (1.10). Each window functional is finite almost surely, and (A.3) bounds its exponential moment. Thus both constants are finite and strictly positive. □

Proof of Corollary 2.1. By (1.10) and (3.4), the conditional law of $(\tau - u/a)/s_u$ given $\mathcal{A}(u)$ converges to the standard normal law. Every successful start is an ordinary crossing. On $\mathcal{A}(u)$ there are successful starts arbitrarily close to S_u, including when the strict continuous-window condition fails at S_u itself. Consequently,

$$\mathcal{A}(u) \cap \{|S_u - \tau| > R\} \subset E_R(u).$$

Dividing the bound in Lemma A.3 by the equivalent in (1.10) gives

$$\lim_{R\to\infty} \limsup_{u\to\infty} \mathbb{P}\{|S_u - \tau| > R \mid \mathcal{A}(u)\} = 0.$$

Since $s_u \to \infty$ and $T_u - S_u$ is the fixed window duration on $\mathcal{A}(u)$, both $(S_u - \tau)/s_u$ and $(T_u - \tau)/s_u$ tend to zero in conditional probability. Slutsky's theorem [3] gives the joint convergence in (2.1).

For $U_u = S_u$ or T_u, let F_u be the conditional distribution function of $(U_u - u/a)/s_u$ given $\mathcal{A}(u)$. Since Φ is continuous, $\sup_x |F_u(x) - \Phi(x)| \to 0$. For finite t, the event $\{U_u \le t\}$ implies $\mathcal{A}(u)$, so

$$\mathbb{P}\{U_u \le t\} = \mathbb{P}\{\mathcal{A}(u)\}F_u((t - u/a)/s_u).$$

Together with (1.10), this proves (2.2). □

Proof of Proposition 2.2. Let $S_a = \sup_{v\ge 0}(W(v) - av)$ and $d = 2a - \kappa = a - c$. By (1.7) and (1.11), $C_{0,0} = \beta\mathbb{E}\{\exp(\kappa \max(S_a, M))\}/(a\kappa)$. The two maxima are independent. Since $M \ge 0$, the exponential tail in (A.1) gives

$$\mathbb{E}\{\exp(\kappa \max(S_a, M)) \mid M\} = \exp(\kappa M) + \kappa\int_M^\infty \exp((\kappa - 2a)x)\mathrm{d}x = \exp(\kappa M) + \frac{\kappa}{d}\exp(-dM).$$

The identity $\kappa d = 2\beta$ gives (2.4). Inserting

$$\mathbb{E}\{\exp(\kappa M)\} = 1 + \kappa\int_0^\infty \exp(\kappa x)H(x)\mathrm{d}x, \qquad \mathbb{E}\{\exp(-dM)\} = 1 - d\int_0^\infty \exp(-dx)H(x)\mathrm{d}x$$

yields (2.3). These integrals are finite by Lemma A.1.

For $u > 0$ let $Y_0(t) = B_0(t) - ct$ and let ϑ_u be its first passage time above u. It agrees almost surely with the first hit of u. Indeed, ordinary Brownian motion has a positive maximum

on every $[0, 1/n]$ almost surely by reflection. Absolute continuity on finite intervals gives the same property with drift $-c$, and the strong Markov property applies at the first hit. Thus on $\{\vartheta_u < \tau\}$ ruin occurs on the ancestral path. On $\{\tau < \vartheta_u\}$, conditioning on the ancestor up to τ gives ruin probability $H(u - Y_0(\tau))$, by independence of the cohort. The two times are independent and τ has a density. Thus the hitting-time transform and conditioning on τ give

$$p_{0,0}(u) = \exp(-\kappa u) + \beta \mathbb{E}\left\{ \int_0^{\vartheta_u} \exp(-\beta t) H(u - Y_0(t)) \mathrm{d}t \right\}. \tag{3.5}$$

For $y < u$, reflection at u followed by the change to drift $-c$ gives the transition density of Y_0 killed at ϑ_u,

$$\exp(-cy - c^2 t/2)\big(\phi_t(y) - \phi_t(2u - y)\big).$$

Multiplying by $\exp(-\beta t)$ and integrating in t as in (A.5) gives the resolvent density

$$R^u_\beta(0, y) = \frac{\exp(-cy)}{a}\big(\exp(-a|y|) - \exp(-a(2u - y))\big).$$

In (3.5) the second term is therefore $\beta \int_{-\infty}^u H(u - y) R^u_\beta(0, y) \mathrm{d}y$. Setting $x = u - y$ yields

$$\exp(\kappa u) p_{0,0}(u) = 1 + \frac{\beta}{a} \int_0^\infty \big(\exp(au + cx - a|u - x|) - \exp(-dx)\big) H(x) \mathrm{d}x. \tag{3.6}$$

The first exponential in (3.6) equals $\exp(\kappa x)$ for $x \le u$ and $\exp(\kappa x - 2a(x - u))$ for $x > u$. Subtraction from (2.3) proves (2.5). At $u = 0$, the ancestor crosses zero arbitrarily soon almost surely, so $p_{0,0}(0) = 1$ and (2.5) follows directly from (2.3). For $\kappa < r < c + \sqrt{c^2 + 4\beta}$, Lemma A.1 gives the uniform error bound

$$0 \le C_{0,0} \exp(-\kappa u) - p_{0,0}(u) \le \frac{\beta D_r}{a(r - \kappa)} \exp(-ru), \qquad u \ge 0.$$

To prove strictness in (2.3), we show $H(1) > 0$ even for singular sibling covariance matrices. There is positive probability that the k_1 initial siblings split at times s_i in disjoint intervals $I_1 < \cdots < I_{k_1}$ in $(0, 1)$ and that the lifetimes of all $k_1 K$ potential children of these siblings exceed two. These are constraints on finitely many independent exponential variables, each in a set of positive probability. Condition on the lifetimes. On this event the Gaussian vector $(W^1_i(s_i))_{1 \le i \le k_1}$ has positive definite covariance. If j is the largest index with $\lambda_j \ne 0$, the last time interval gives, with $s_0 = 0$,

$$\mathrm{Var}\left(\sum_i \lambda_i W^1_i(s_i) \,\middle|\, (s_i)_i \right) \ge \lambda_j^2 (s_j - s_{j-1}) > 0.$$

Indeed, the increment of coordinate j on that interval is independent of the vector up to s_{j-1}. Hence all initial siblings can split at positions exceeding $2c + 2$ with positive conditional probability. For each newborn sibling group the Gaussian vector of displacements at time 2 has positive probability of belonging to $(-1, 1)^k$, where k is its size. Indeed it is a linear image of a standard Gaussian vector, and the inverse image of this box contains a neighbourhood of zero. These vectors are independent across births and of the initial vector. On the intersection all particles at time 2 lie above $2c + 1$. Thus $H(x) \ge H(1) > 0$ on $(0, 1)$, which gives $C_{0,0} > 1$. □

Proof of Corollary 2.3. Let f be continuous with $f(s) \to -\infty$ as $|s| \to \infty$. Fix $0 \le H < \infty$. For durations at most H and meshes at most one, a continuous window starting at zero or a sampled window starting at the first nonnegative lattice point has minimum at least $B = \min_{[0, H+1]} f$. Choose $R > 1$ such that $f(s) < B$ for $|s| > R$. Since each window contains its start, all relevant suprema are attained at starts in $[-R, R]$. Write ω_f for the modulus of continuity on $[-R - 1, R + H + 1]$.

For $h, h' \le H$, the intervals $[s, s + h]$ and $[s, s + h']$ have Hausdorff distance $|h - h'|$. Their minima differ by at most $\omega_f(|h - h'|)$ for $|s| \le R$, giving the same bound for $|\Gamma_h(f) - \Gamma_{h'}(f)|$. For (2.6) take $H > h$ and n large enough that $m_n \delta_n \le H$ and $\delta_n \le 1$. From a maximising continuous start move to a grid start at distance at most δ_n. The resulting sampled window

and the original interval have Hausdorff distance at most $2\delta_n + |m_n\delta_n - h|$. Conversely, at a maximising grid start use the continuous window of length h with that same start. Both comparisons take place in $[-R-1, R+H+1]$ and give

$$\sup_{0\le\theta<\delta_n} |V_{\delta_n,m_n,\theta}(f) - \Gamma_h(f)| \le \omega_f(2\delta_n + |m_n\delta_n - h|) \longrightarrow 0.$$

Apply the two uniform comparisons to Z, using (A.2). The exponentials are bounded by $\exp(\kappa \sup_s Z(s))$, which is integrable by (A.3). Dominated convergence gives continuity in h. For the grid, substitute $\theta = \delta_n v$ in (1.12) and apply the same bound on the product space with $0 \le v < 1$ to obtain (2.6).

For ordinary ruin let U be uniform on $(0,1)$ independently of Z. By independence of S_a and M, $\mathbb{P}\{S_a > M\} = \mathbb{E}\{\exp(-2aM)\} > 0$. At each fixed $v \ge 0$, $W(v) - av < S_a$ almost surely, since the future increments have supremum distributed as S_a and $\mathbb{P}\{S_a > 0\} = 1$. Conditioning on U and taking a countable intersection, every negative grid point has value strictly below S_a almost surely. On $\{S_a > M\}$ the nonnegative grid values are also strictly smaller. The grid maximum is attained by (A.2), so it is strictly smaller than $\sup_s Z(s)$ on this event. Elsewhere it is at most $\sup_s Z(s)$. Taking expectations proves the strict inequality in (2.7). □

Appendix A. Technical estimates

For $d > 0$ put $S_d = \sup_{s\ge0}(W(s) - ds)$, with W as in (1.7). The Brownian maximum identity [4] gives

$$\mathbb{P}\{S_d > x\} = \exp(-2dx), \qquad x \ge 0. \tag{A.1}$$

Lemma A.1. *For the newborn cohort in (1.6), put $\gamma_* = c + \sqrt{c^2 + 4\beta}$. For every $0 < r < \gamma_*$ there exists $D_r < \infty$, depending only on c, β, r, such that*

$$\mathbb{P}\{M > u\} \le D_r \exp(-ru), \qquad u \ge 0.$$

In particular $\mathbb{E}\{\exp(\kappa M)\}$ is bounded uniformly over the admissible offspring and covariance sequences.

Proof. Retain the first two initial siblings, each with weight $1/2$. At each split of a retained particle of weight w, retain its first two children, each with weight $w/2$. All labels in this binary tree are born almost surely, since every split produces at least two children. Write $\mathcal{B}(t)$ for its living particles and $w_v = 2^{-g}$ for the weight of a particle at depth g, counting the initial siblings at depth one. The living weights sum to one. Their weighted position

$$\overline{X}(t) = \sum_{v\in\mathcal{B}(t)} w_v X_v(t)$$

is continuous at splits, and $L(t) \le \overline{X}(t) - ct$.

Condition on the lifetimes of all potential particles, denoting their σ-field by $\mathcal{G}$. This fixes every birth time b_v and the chronological assignment of the covariance matrices. If W_v is the Brownian coordinate assigned at the birth of v, conservation of descendant weights gives

$$\overline{X}(t) = \sum_{v \text{ retained}} w_v W_v\big((t - b_v)_+ \wedge \zeta_v\big).$$

Only finitely many terms contribute on each compact time interval. Conditionally on $\mathcal{G}$ this is a centred Gaussian process with independent increments. Indeed, distinct birth events use independent vectors. Within a single vector all siblings have the same birth time, so disjoint time intervals use disjoint increments of the vector, also when an individual coordinate stops at its splitting time.

Let $V(t) = \mathrm{Var}(\overline{X}(t) \mid \mathcal{G})$. At a time between splits group the living retained particles by their birth event e, including the initial birth at time zero. Write $J_e(t)$ for the living group and R^e

for its sibling correlation matrix. The variance rate on this interval satisfies

$$0 \le V'(t) = \sum_e \sum_{i,j \in J_e(t)} w_i w_j R^e_{ij} \le \sum_e \left(\sum_{i \in J_e(t)} w_i \right)^2 \le 2 \sum_{v \in \mathcal{B}(t)} w_v^2.$$

The last inequality uses $|J_e(t)| \le 2$. The sum of squared weights starts at $1/2$ and decreases by $w^2/2$ at a retained split of weight w. Thus $V(t) \le t$. Integrating the same variance bound over the lifetime of each particle gives

$$V(t) \le J_\infty := 2 \sum_{g \ge 1} 4^{-g} \sum_{v \text{ retained at depth } g} \zeta_v.$$

The labels in this last sum form a deterministic tree, with 2^g independent $\mathrm{Exp}(\beta)$ lifetimes at depth g and independence across depths. Hence for $0 \le \alpha < 2\beta$,

$$\mathbb{E}\{\exp(\alpha J_\infty)\} = \prod_{g \ge 1} \left(1 - 2\alpha 4^{-g}/\beta\right)^{-2^g} < \infty.$$

Indeed the largest $2\alpha 4^{-g}/\beta$ is $\alpha/(2\beta) < 1$, and $\sum_{g \ge 1} 2^g 4^{-g} < \infty$ bounds the logarithm of the product up to a constant.

For $r \in (0, \gamma_*)$ put $\alpha_r = (r^2/2 - rc)_+ < 2\beta$. Conditionally on $\mathcal{G}$, the process $\mathcal{M}_r(t) = \exp(r\overline{X}(t) - r^2 V(t)/2)$ is a mean-one martingale in the natural filtration of $\overline{X}$. The two bounds on V give

$$r^2 V(t)/2 - rct \le (r^2/2 - rc) V(t) \le \alpha_r J_\infty.$$

Thus a crossing by L above u entails a crossing by $\mathcal{M}_r$ above $\exp(ru - \alpha_r J_\infty)$. Apply the nonnegative martingale maximal inequality [11] conditionally on $\mathcal{G}$ on $[0, T]$ and let $T \to \infty$. Averaging then gives the stated bound with $D_r = \mathbb{E}\{\exp(\alpha_r J_\infty)\}$. Since $\kappa < \gamma_*$, any $r \in (\kappa, \gamma_*)$ yields

$$\mathbb{E}\{\exp(\kappa M)\} = 1 + \kappa \int_0^\infty \exp(\kappa u) \mathbb{P}\{M > u\} \mathrm{d}u \le 1 + \frac{\kappa D_r}{r - \kappa}.$$

□

Choose $0 < \varepsilon < \min\{c, (a-c)/2\}$ and put $M_\varepsilon = \sup_{s \ge 0}(L(s) + \varepsilon s)$. Since $\kappa(a - c) = 2\beta$,

$$\kappa^2/2 - \kappa(c - \varepsilon) = \beta + \kappa\varepsilon < 2\beta, \qquad \kappa < 2(a - \varepsilon).$$

The first inequality places κ strictly below the tail exponent in Lemma A.1 with drift $c - \varepsilon$. Tail integration with an exponent strictly between the two gives a uniform bound on $\mathbb{E}\{\exp(\kappa M_\varepsilon)\}$. The second inequality and (A.1) give $\mathbb{E}\{\exp(\kappa S_{a-\varepsilon})\} < \infty$. By (1.7),

$$\sup_{s \in \mathbb{R}} (Z(s) + \varepsilon |s|) = \max\{S_{a-\varepsilon}, M_\varepsilon\}.$$

Consequently,

$$L(s) \to -\infty \quad (s \to \infty), \qquad Z(s) \to -\infty \quad (|s| \to \infty) \qquad \text{almost surely,} \tag{A.2}$$

and, uniformly over the environments,

$$\mathbb{E}\left\{ \exp\left(\kappa \sup_{s \in \mathbb{R}} Z(s)\right) \right\} = \mathbb{E}\{\exp(\kappa \max\{S_a, M\})\} < \infty. \tag{A.3}$$

The same choice of ε gives, uniformly in the environment,

$$\begin{aligned} \exp(\kappa Z(s)) &\le A_\varepsilon \exp(-\kappa\varepsilon|s|), \qquad && s \in \mathbb{R}, \\ A_\varepsilon &= \exp\big(\kappa \max\{S_{a-\varepsilon}, M_\varepsilon\}\big), && \mathbb{E}\{A_\varepsilon\} < \infty. \end{aligned} \tag{A.4}$$

For continuous observation the moving minimum in (1.8) is continuous in its start by uniform continuity on compact intervals. It is bounded above by the value at the start, which tends to $-\infty$ at either end. It therefore attains a finite maximum. The same argument on the shifted lattice gives attainment in (1.9). On continuous path space the minimum over $[0, h]$ can be taken over a fixed countable dense subset (a singleton if $h = 0$). Continuity in the start

also permits a countable dense set of starts, both on $\mathbb{R}$ and on each half-open block. This gives measurability of the continuous functionals and block events. In (1.9) the countable supremum of finite minima of evaluations is jointly measurable in the path and its phase. Thus the exponentials defining the constants in (1.11) and (1.12) are strictly positive and integrable by (A.3).

Lemma A.2. *Let ϕ_t denote the centred normal density of variance t, write $\phi = \phi_1$, and put $\nu_z(\mathrm{d}t) = a\phi_t(z - at)\mathrm{d}t$ on $(0, \infty)$. For $z \in \mathbb{R}$,*

$$\int_0^\infty \phi_t(z - at)\mathrm{d}t = a^{-1}\exp(az - a|z|) \le a^{-1}. \tag{A.5}$$

For either observation rule, uniformly over the admissible environments,

$$\sup_{u>0} \exp(\kappa u)\mathbb{P}\{\mathcal{A}(u)\} < \infty, \qquad \lim_{\Lambda\to\infty} \limsup_{u\to\infty} \exp(\kappa u)\mathbb{P}\{\mathcal{A}(u),\ |\tau - u/a| > \Lambda s_u\} = 0. \tag{A.6}$$

Proof. Writing $\phi_t(z-at) = \exp(az-a^2t/2)\phi_t(z)$ in the integral and using the Gaussian integral $\int_0^\infty t^{-1/2}\exp(-(a^2t + z^2/t)/2)\mathrm{d}t = \sqrt{2\pi}\exp(-a|z|)/a$ gives (A.5).
Every successful window contains a time with $Y > u$. We bound this ordinary crossing according to the deficit $x = u - Y(\tau)$ at the first split. Multiplying the joint density of (τ, x) by $\exp(\kappa u)$ gives

$$\beta\exp(\kappa u - \beta t)\phi_t(u + ct - x)\mathrm{d}t\mathrm{d}x = \frac{\beta}{a}\exp(\kappa x)\nu_{u-x}(\mathrm{d}t)\mathrm{d}x, \tag{A.7}$$

since $a^2 = c^2 + 2\beta$ and $\kappa = a + c$. Conditional on (τ, x), the probability of a crossing at or after the split is $H(x)$, with $H(x) = 1$ for $x < 0$. It remains to count crossings before the split whose endpoint is $u - x \le u$. For $x \ge 0$, reflection at u for standard Brownian motion gives the endpoint density $\phi_t(u + x)$ on paths that reached u by t. Introducing drift $-c$ multiplies this density by $\exp(-c(u - x) - c^2t/2)$ [4]. Thus, for every Borel $J \subset (0, \infty)$,

$$\begin{aligned}\exp(\kappa u)\mathbb{P}\{\mathcal{A}(u),\ \tau \in J\} \le{}& \frac{\beta}{a}\int_{\mathbb{R}} \exp(\kappa x)H(x)\nu_{u-x}(J)\mathrm{d}x \\ &+ \frac{\beta}{a}\int_0^\infty \exp(-(a - c)x)\nu_{u+x}(J)\mathrm{d}x.\end{aligned}$$

Both kernels have mass at most one. For any fixed $\kappa < r < c + \sqrt{c^2 + 4\beta}$, Lemma A.1 bounds the first weight on $x \ge 0$ by $D_r\exp(-(r - \kappa)x)$. On $x < 0$ it equals $\exp(\kappa x)$. These integrable bounds and $a > c$ prove the first assertion of (A.6).
For each fixed $x \in \mathbb{R}$, ν_{u+x} has mass one once $u > |x|$. Under $y = (t - u/a)/s_u$ its density on $t = u/a + ys_u > 0$ is

$$\frac{as_u}{\sqrt{t}}\,\phi\!\left(\frac{x - ays_u}{\sqrt{t}}\right).$$

Extend it by zero elsewhere. Since $as_u = \sqrt{u/a}$, this density converges pointwise to $\phi(y)$ and hence in L^1 by Scheffé's lemma [21]. The same holds with $-x$ in place of x. For $J = \{t > 0 : |t - u/a| > \Lambda s_u\}$ both kernel masses thus tend to $2\Phi(-\Lambda)$. Replacing H by the common bound above and applying dominated convergence bounds the limit superior, uniformly in the environment, by a constant times $2\Phi(-\Lambda)$. Letting $\Lambda \to \infty$ proves (A.6). □

Lemma A.3. *Let*

$$E_R(u) = \{\exists t \ge 0 : |t - \tau| > R,\ Y(t) > u\}.$$

Then, uniformly over all admissible environments,

$$\lim_{R\to\infty} \limsup_{u\to\infty} \exp(\kappa u)\mathbb{P}\{E_R(u)\} = 0. \tag{A.8}$$

Proof. For a crossing before $\tau - R$, the ancestral hitting time $\vartheta_u = \inf\{s \geq 0 : B_0(s) - cs > u\}$ satisfies $\vartheta_u + R < \tau$. Independence and the Brownian hitting-time transform give

$$\mathbb{P}\{\tau > \vartheta_u + R\} = \exp(-\beta R)\mathbb{E}\{\exp(-\beta\vartheta_u)\} = \exp(-\kappa u - \beta R).$$

For a crossing after $\tau + R$, condition on the deficit $x = u - Y(\tau)$. There must be an $s > R$ with $L(s) > x$. With ε from (A.4) and $M_\varepsilon = \sup_{s\geq 0}(L(s) + \varepsilon s)$, this implies $M_\varepsilon > x + \varepsilon R$. By independence of the cohort, (A.7) and the mass bound in (A.5), we obtain

$$\begin{aligned}\exp(\kappa u)\mathbb{P}\{E_R(u)\} &\leq \exp(-\beta R) + \frac{\beta}{a}\int_{\mathbb{R}} \exp(\kappa x)\mathbb{P}\{M_\varepsilon > x + \varepsilon R\}\mathrm{d}x \\ &= \exp(-\beta R) + \frac{\beta}{a\kappa}\exp(-\kappa\varepsilon R)\mathbb{E}\{\exp(\kappa M_\varepsilon)\}.\end{aligned}$$

The equality follows by integrating first over $x < M_\varepsilon - \varepsilon R$. The expectation is bounded uniformly by Lemma A.1 with drift $c - \varepsilon$, so the bound tends to zero uniformly in u and the environment. □

Remark (Other environments). Suppose that $\mathcal{E}$ satisfies (1.1) almost surely with a deterministic K and is independent of the K-dimensional Brownian drivers, B_0, and the lifetimes. The entries of $\mathcal{E}$ may be dependent. The common array of clocks and drivers constructs the process for every realisation of $\mathcal{E}$. On a finite time interval the construction uses finitely many splits, so its evaluations are measurable in the environment, clocks and drivers. Conditionally on $\mathcal{E}$, Theorem 1.1 and Corollary 2.1 apply with the constant $C(\mathcal{E})$ of the chosen rule.
For the unconditional law, put $y_u(t) = (t - u/a)/s_u$ and take a bounded continuous f. By (3.4),

$$\exp(\kappa u)\mathbb{E}\big\{f(y_u(\tau))\mathbf{1}_{\mathcal{A}(u)} \mid \mathcal{E}\big\} \longrightarrow C(\mathcal{E})\mathbb{E}\{f(G)\} \qquad \text{almost surely.}$$

The uniform bound in (A.6) provides a deterministic majorant proportional to $\|f\|_\infty$. Averaging gives the same limit with $C = \mathbb{E}\{C(\mathcal{E})\}$. The choice $f = 1$ gives the ruin equivalent with $0 < C < \infty$. Averaging the uniform bound in Lemma A.3 transfers the conditional limit given ruin from τ to S_u, T_u as in the proof of Corollary 2.1.
Alternatively, assign (k_{g+1}, A_{g+1}) to the split of each parent at generation g, counting the ancestor at generation zero. The population size after j splits may now be random. Coupling to the full K-ary tree with the same lifetimes gives nonexplosion. Relative to the first split, the new cohort starts with the first sibling matrix, and its subsequent evolution uses only descendant clocks and fresh Brownian drivers. Its law is therefore independent of the time and position of the first split. For a fixed environment, conditioning on the clocks fixes the matrix at each birth, and the first two children at each split still form the binary tree in Lemma A.1. These are the only properties of the assignment used in the proof of Theorem 1.1.

Appendix B. A spectral interpretation

Set next

$$F(t) = \exp(\kappa Z(t)), \qquad I(f) = \int_{\mathbb{R}} f(s)\mathrm{d}s, \qquad b = \mathbb{E}\{I(F)\} \tag{B.1}$$

and hence in view of (A.4) $0 < b < \infty$. Write $\mathbb{P}_F$ for the law of F and choose a path and an origin (F, U) with joint law

$$\mathbb{P}_*(\mathrm{d}f, \mathrm{d}s) = \frac{f(s)}{b}\mathbb{P}_F(\mathrm{d}f)\mathrm{d}s, \qquad \widehat{F}(t) = \frac{F(U+t)}{F(U)}. \tag{B.2}$$

This change of origin gives $\mathbb{E}_*\Big\{\widehat{F}(t)\Big\} = 1$ for every t, by integration in s. For the general tilt–shift representation see [13].
Let $\{f_i\}$ be a Poisson point process of profiles with intensity

$$\nu_F(A) = \frac{1}{b}\int_{\mathbb{R}}\int_0^\infty \mathbb{P}\{rF(s+\cdot) \in A\}\frac{\mathrm{d}r}{r^2}\mathrm{d}s = \mathbb{E}_*\left\{\int_0^\infty \mathbf{1}_{\{r\widehat{F}\in A\}}\frac{\mathrm{d}r}{r^2}\right\}. \tag{B.3}$$

In the last integral, substituting $r = F(s)y$ in (B.2) cancels the factor $F(s)$. Translation of s leaves ν_F invariant and scaling of r gives $\nu_F(yA) = y^{-1}\nu_F(A)$ for $y > 0$. At each site, radial integration gives $\nu_F\{f : f(t) > x\} = 1/x$ for $x > 0$. For each compact interval J, the mass of $\{f : \sup_{t\in J} f(t) > x\}$ is finite by (A.4), since $\sup_{t\in J} F(s+t) \le A_\varepsilon \exp(-\kappa\varepsilon \operatorname{dist}(-s, J))$. The Poisson void probability now gives $\mathbb{P}\{X(t) \le x\} = \exp(-1/x)$ for $X(t) = \sup_i f_i(t)$. Translation invariance gives stationarity, and radial homogeneity gives max-stability. On a compact interval the maxima of the finitely many profiles exceeding $1/n$ there approximate X uniformly to within $1/n$ (with the empty maximum taken as zero). Thus X has continuous paths.

For a positive continuous path f define

$$\mathcal{P}_{0,h}(f) = \sup_{s\in\mathbb{R}} \inf_{0\le r\le h} f(s+r), \quad \mathcal{P}_{\delta,m}(f) = \frac{1}{\delta}\int_0^\delta \sup_{j\in\mathbb{Z}} \min_{\ell\in\{0,\dots,m\}} f((j+\ell)\delta - \theta)\mathrm{d}\theta.$$

These functionals are homogeneous of degree one and translation invariant, since a translation on the grid changes the phase modulo δ. For either observation rule $\star$, (1.11) and (1.12) give

$$C_\star = \frac{\beta}{a\kappa}\mathbb{E}\{\mathcal{P}_\star(F)\} = \frac{\beta b}{a\kappa}\mathbb{E}_*\left\{\frac{\mathcal{P}_\star(\widehat{F})}{I(\widehat{F})}\right\}. \tag{B.4}$$

Indeed the ratio is unchanged by the shift and rescaling in (B.2), and F has marginal law $I(f)\mathbb{P}_F(\mathrm{d}f)/b$ under $\mathbb{P}_*$.

For $T > 0$ let $\Psi_{T,h}(f) = \sup_{0\le s\le T} \inf_{0\le r\le h} f(s+r)$. Tonelli's theorem [12] and radial integration in the two expressions in (B.3) give, for $x > 0$,

$$x\,\nu_F\{f : \Psi_{T,h}(f) > x\} = \mathbb{E}_*\left\{\Psi_{T,h}(\widehat{F})\right\} = \frac{1}{b}\int_{\mathbb{R}} \mathbb{E}\{\Psi_{T,h}(F(s+\cdot))\}\mathrm{d}s = \frac{\mathcal{H}_T}{b}.$$

The last equality uses the continuous block integral from the proof of Theorem 1.1. The moving infimum is continuous, so the supremum over $[0, T)$ equals that over $[0, T]$. Its block limit and (B.4) yield

$$\lim_{T\to\infty} \frac{1}{T}\mathbb{E}_*\left\{\Psi_{T,h}(\widehat{F})\right\} = \frac{a\kappa}{\beta b}C_{0,h}. \tag{B.5}$$

This is the limiting mean number per unit time of individual profiles that exceed level one throughout some window of length h. For $h = 0$, $\Psi_{T,0}(X) = \sup_i \Psi_{T,0}(f_i)$ and (B.5) gives the usual Pickands constant [7]. For $h > 0$ one has $\Psi_{T,h}(X) \ge \sup_i \Psi_{T,h}(f_i)$ by monotonicity. Different profiles can sustain the exceedance on different parts of one window.

References

[1] L. Bai. Asymptotics of Parisian ruin of Brownian motion risk model over an infinite-time horizon. *Scandinavian Actuarial Journal* **2018** (2018), no. 6, 514–528.

[2] J. Berestycki, É. Brunet, S. C. Harris, and P. Miłoś. Branching Brownian motion with absorption and the all-time minimum of branching Brownian motion with drift. *Journal of Functional Analysis* **273** (2017), 2107–2143.

[3] P. Billingsley. *Convergence of Probability Measures*. Second edition. Wiley Series in Probability and Statistics. Wiley, New York, 1999.

[4] A. N. Borodin and P. Salminen. *Handbook of Brownian Motion—Facts and Formulae*. Second edition. Birkhäuser, Basel, 2002.

[5] I. Czarna and Z. Palmowski. Ruin probability with Parisian delay for a spectrally negative Lévy risk process. *Journal of Applied Probability* **48** (2011), no. 4, 984–1002.

[6] I. Czarna, Z. Palmowski, and P. Świątek. Discrete time ruin probability with Parisian delay. *Scandinavian Actuarial Journal* **2017** (2017), no. 10, 854–869.

[7] K. Dębicki and E. Hashorva. Approximation of supremum of max-stable stationary processes & Pickands constants. *Journal of Theoretical Probability* **33** (2020), no. 1, 444–464.

[8] K. Dębicki, E. Hashorva, and L. Ji. Parisian ruin of self-similar Gaussian risk processes. *Journal of Applied Probability* **52** (2015), no. 3, 688–702.
[9] K. Dębicki, E. Hashorva, and L. Ji. Parisian ruin over a finite-time horizon. *Science China Mathematics* **59** (2016), no. 3, 557–572.
[10] K. Dębicki, P. Ievlev, and N. Kriukov. Extremes of Brownian Decision Trees. arXiv:2512.05248 (2025).
[11] J. L. Doob. *Stochastic Processes*. Wiley, New York, 1953.
[12] G. B. Folland. *Real Analysis: Modern Techniques and Their Applications*. Second edition. Wiley, New York, 1999.
[13] E. Hashorva. Representations of max-stable processes via exponential tilting. *Stochastic Processes and their Applications* **128** (2018), no. 9, 2952–2978.
[14] P. Ievlev. Parisian ruin with power-asymmetric variance near the optimal point with application to many-inputs proportional reinsurance. *Stochastic Models* **40** (2024), no. 3, 518–535.
[15] G. Jasnovidov. Approximation of ruin probability and ruin time in discrete Brownian risk models. *Scandinavian Actuarial Journal* **2020** (2020), no. 8, 718–735. arXiv:2001.10311.
[16] I. Karatzas and S. E. Shreve. *Brownian Motion and Stochastic Calculus*. Second edition. Graduate Texts in Mathematics, volume 113. Springer, New York, 1991.
[17] R. Loeffen, I. Czarna, and Z. Palmowski. Parisian ruin probability for spectrally negative Lévy processes. *Bernoulli* **19** (2013), no. 2, 599–609. arXiv:1102.4055.
[18] S. M. Novikov. Parisian ruin of locally self-similar Gaussian processes. *Scandinavian Actuarial Journal*, published online 28 May 2026.
[19] Z. Palmowski, L. Ramsden, and A. D. Papaioannou. Parisian ruin for the dual risk process in discrete-time. *European Actuarial Journal* **8** (2018), no. 1, 197–214.
[20] G. Popivoda. Parisian ruin of Gaussian processes with random premium income. *Stochastics* **97** (2025), no. 5, 645–666.
[21] H. Scheffé. A useful convergence theorem for probability distributions. *Annals of Mathematical Statistics* **18** (1947), 434–438.

Enkelejd Hashorva, Department of Actuarial Science, University of Lausanne, UNIL-Dorigny, 1015 Lausanne, Switzerland
Email address: Enkelejd.Hashorva@unil.ch

Goran Popivoda, Faculty of Science and Mathematics, University of Montenegro, Džordža Vašingtona bb, Podgorica, Montenegro
Email address: goranp@ucg.ac.me

Svyatoslav Novikov, Department of Actuarial Science, University of Lausanne, UNIL-Dorigny, 1015 Lausanne, Switzerland
Email address: Svyatoslav.Novikov@unil.ch

Timofei Shashkov, Department of Actuarial Science, University of Lausanne, UNIL-Dorigny, 1015 Lausanne, Switzerland
Email address: Timofei.Shashkov@unil.ch